

NTRK3 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes NTRK3 gene fusions from the msk_impact_50k_2026 study, covering 58 fusion events. 55/58 fusions (94.8%) were in-frame. 55/58 fusions (94.8%) retained the Protein tyrosine and serine/threonine kinase domain. Domain retention was tested with Fisher's exact test ($p=0.0695489$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for NTRK3.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 58 analyzed fusion events, identifying 4 distinct fusion partner genes. The most frequent partner was ETV6, observed in 55 events.

Domain retention

Protein tyrosine and serine/threonine kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 54/55 (98.2%) of in-frame fusions retained the domain, $p=0.0695489$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

Domain disruption

Domain-disruption analysis was skipped: NTRK3 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 57 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 155 aa (permutation-corrected $p=0.036963$, statistically significant at $\alpha=0.05$). This is -5 aa from the nearest configured domain boundary at 160 aa.

Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: NTRK3 has no group_field configured; confidence-stats analysis was skipped.

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

Joint partner

Joint partner reported: NTRK3 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ETV6	55	0.9482758620689655	0.20043213737826426	None	0.0	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.47798684709322253	1
DOT1L	1	0.017241379310344827	0.20043213737826426	None	1.0	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2932577830538136	2
ZNF710	1	0.017241379310344827	0.20043213737826426	None	0.11159235668789813	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.10288471662979178	3
VPS39	1	0.017241379310344827	0.20043213737826426	None	None	None	['domain_retention', 'recurrence']	0.10050990570485364	4

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
155	0	55	1	1	0.0	0.03508771929824562	1.4548448600085102
466	14	41	2	0	0.0	0.07518796992481205	1.1238516409670856
472	15	40	2	0	0.0	0.08521303258145364	1.069493978644493

domain_retention tables

frame_domain_contingency_table

54	1
1	1

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein tyrosine and serine/threonine kinase	57	55	0	2	0.0

frequency tables

Partner_gene_counts

Partner_gene	Event_count
DOT1L	1
ETV6	55
VPS39	1
ZNF710	1

Per-event results (results.tsv)

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-001 3664-T02-I M5-1	P-0013 664-T0 2-IM5		ETV 6-NT RK3	ETV 6	in-fr ame	thre e_ _pr im e	885581 91	13	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q26.1)(chr12:g.12032077::chr15:g.88558191) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-006 1265-T01-I M7-2	P-0061 265-T0 1-IM7		ETV 6-NT RK3	ETV 6	in-fr ame	thre e_ _pr im e	885541 22	13	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12033175::chr15:g.88554122) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-005 8408-T01-I M6-3	P-0058 408-T0 1-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thre e_ _pr im e	884921 56	14	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12035760::chr15:g.88492156) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-005 6156-T02-I M7-4	P-0056 156-T0 2-IM7		ETV 6-NT RK3	ETV 6	in-fr ame	thre e_ _pr im e	886242 74	12	466	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12011957::chr15:g.88624274) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-005 5703-T01-I M6-5	P-0055 703-T0 1-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thr ee_ _pr im e	885059 80	14	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 24785::chr15:g.88505980) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-004 2503-T01-I M6-6	P-0042 503-T0 1-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thr ee_ _pr im e	885560 67	13	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 32802::chr15:g.88556067) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-004 0355-T01-I M6-7	P-0040 355-T0 1-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thr ee_ _pr im e	884971 91	14	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 36111::chr15:g.88497191) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 7610-T01-I M6-8	P-0037 610-T0 1-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thr ee_ _pr im e	885945 32	13	466	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) rearrangement: t(12;15)(p13.2 ;q25.3)(chr12:g.12014663::chr 15:g.88594532) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 5767-T06-I M7-9	P-0035 767-T0 6-IM7		ETV 6-NT RK3	ETV 6	in-fr ame	thr ee_ _pr im e	885057 87	14	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 29143::chr15:g.88505787) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 5767-T04-I M6-10	P-0035 767-T0 4-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thr ee_ _pr im e	885057 87	14	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 29143::chr15:g.88505787) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 3921-T05-I M7-11	P-0033 921-T0 5-IM7		ETV 6-NT RK3	ETV 6	in-fr ame	thr ee_ _pr im e	885165 34	14	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 31161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-003 3921-T04-I M6-12	P-0033 921-T0 4-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thr ee_ _pr im e	885165 34	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 31161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 3921-T02-I M6-13	P-0033 921-T0 2-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thr ee_ _pr im e	885165 34	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 31161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 2148-T04-I M6-14	P-0032 148-T0 4-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thr ee_ _pr im e	885224 37	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 37217::chr15:g.88522437) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-006 4964-T01-I M7-15	P-0064 964-T0 1-IM7		ETV 6-NT RK3	ETV 6	in-fr ame	thr ee_ _pr im e	885646 82	13	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion: t(12; 15)(p13.2;q25.3)(chr12:g.120 32310::chr15:g.88564682) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 3664-T04-I M6-16	P-0013 664-T0 4-IM6		ETV 6-NT RK3	ETV 6	in-fr ame	thr ee_ _pr im e	885581 91	13	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032077::chr15:g.88 558191) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-005 8230-T02-I M7-19	P-0058 230-T0 2-IM7		NTR K3-D OT1 L	DOT 1L	out- of-fr ame	five _pr im e	886832 44	6	155	True	lost	0.0	False		Prote in tyr osine and s erine /thre onine kinas e		NTRK3 (NM_001012338) - DOT1L (NM_032482) rearrangement: t(15;19)(q25.3 ;p13.3)(chr15:g.88683244::chr 19:g.2227604) Protein Fusion: out of frame {NTRK3:DOT1L}	NA
EVT-P-001 3196-T03-I M6-20	P-0013 196-T0 3-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thr ee_ _pr im e	885013 57	14	529	True	retain ed	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3) (chr12:g.12036072::chr15:g.8 8501357) Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002978-T06-I M6-21	P-0002978-T06-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88506571	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0010668-T01-I M5-22	P-0010668-T01-IM5		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88513282	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029109::chr15:g.88513282) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0014872-T02-I M6-23	P-0014872-T02-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88548029	13	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028946::chr15:g.88548029) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0014872-T03-I M6-24	P-0014872-T03-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88548029	13	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028946::chr15:g.88548029) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0013196-T04-I M6-25	P-0013196-T04-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88501357	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0016507-T01-I M6-26	P-0016507-T01-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88488820	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 6507-T02-IM6-27	P-0016 507-T02-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88488820	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-14 fused with NTRK3 exons 6-20 including the kinase domain): t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 6507-T03-IM6-28	P-0016 507-T03-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88488820	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20: t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 6507-T04-IM6-29	P-0016 507-T04-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88488820	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 fused to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-000 2671-T01-IM3-30	P-0002 671-T01-IM3		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88505784	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-001 8060-T01-IM6-31	P-0018 060-T01-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88514531	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037::chr15:g.88514531) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-002 1974-T01-IM6-32	P-0021 974-T01-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88550500	13	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 fused to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028417::chr15:g.88550500) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 3196-T02-I M6-33	P-0013 196-T0 2-IM6		NTR K3-E TV6	ETV 6	in-frame	threep _prime	885013 57	14	529	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-002 5507-T01-I M6-34	P-0025 507-T0 1-IM6		NTR K3-E TV6	ETV 6	in-frame	threep _prime	885445 73	13	529	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12024078::chr15:g.88544573) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-002 5987-T01-I M6-35	P-0025 987-T0 1-IM6		NTR K3-E TV6	ETV 6	in-frame	threep _prime	886131 14	13	466	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12028942::chr15:g.88613114) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-001 0562-T01-I M5-36	P-0010 562-T0 1-IM5		NTR K3-E TV6	ETV 6	in-frame	threep _prime	885816 58	13	466	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 14-20): t(12;15)(p13.2;q25.1)(chr12:g.12028903::chr15:g.88581658) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-002 6390-T01-I M6-37	P-0026 390-T0 1-IM6		NTR K3-E TV6	ETV 6	in-frame	threep _prime	886511 31	12	466	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12016863::chr15:g.88651131) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-002 6696-T01-I M6-38	P-0026 696-T0 1-IM6		NTR K3-E TV6	ETV 6	in-frame	threep _prime	886579 25	12	466	True	retained	1.0	False	Protein tyrosine and serine/ threonine kinase			ETV6 (NM_001012338) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20):t(12;15)(p13.2;q25.3)(chr12:g.12007967::chr15:g.88657925) Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 7006-T01-I M6-39	P-0027 006-T0 1-IM6		NTR K3-E TV6	ETV 6	in-frame	threonine_prime	88512343	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12032352::chr15:g.88512343) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-002 7006-T03-I M6-40	P-0027 006-T0 3-IM6		NTR K3-E TV6	ETV 6	in-frame	threonine_prime	88512343	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037::chr15:g.88514531) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-002 7006-T04-I M6-41	P-0027 006-T0 4-IM6		NTR K3-E TV6	ETV 6	in-frame	threonine_prime	88512343	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352::chr15:g.88512343) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-000 2978-T07-I M6-42	P-0002 978-T0 7-IM6		NTR K3-E TV6	ETV 6	in-frame	threonine_prime	88506571	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-000 2978-T02-I M5-43	P-0002 978-T0 2-IM5		NTR K3-E TV6	ETV 6	in-frame	threonine_prime	88506571	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-002 6286-T01-I M6-44	P-0026 286-T0 1-IM6		NTR K3-E TV6	ETV 6	in-frame	threonine_prime	88611476	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12030395::chr15:g.88611476) Protein Fusion: in frame {ETV6:NTRK3}	NA

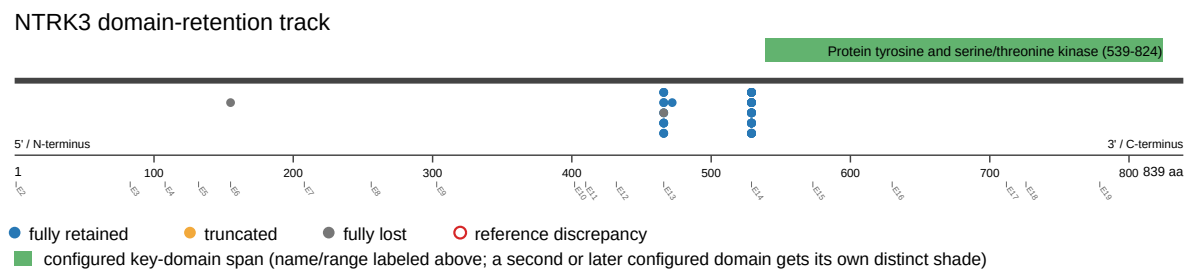
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002978-T03-IM5-45	P-0002978-T03-IM5		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88506571	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1 to 5 fused to NTRK3 exons 15 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0010264-T01-IM5-46	P-0010264-T01-IM5		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88588255	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1-4 fused in frame with NTRK3 exons 13-20): t(12;15)(p13.2;q25.3)(chr12:g.12014962::chr15:g.88588255) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0007911-T02-IM5-47	P-0007911-T02-IM5		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88592430	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion (ETV6 exons 1 to 4 fused with NTRK3 exons 13-20): t(12;15)(p13.2;q25.3)(chr12:g.12007142::chr15:g.88592430) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0033921-T03-IM6-48	P-0033921-T03-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88516534	14	529	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0007772-T03-IM5-49	P-0007772-T03-IM5		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88600844	13	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0007772-T03-IM5-50	P-0007772-T03-IM5		NTRK3-ETV6	ETV6	in-frame	five_prime	88600844	13	466	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		ETV6 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844) Protein fusion: in frame (NTRK3-ETV6)	NA

event_id	sampl_e_id	pa_tie_n_t_id	fusio_n_name	part_ner_ge_ne	fra_me_st_atu_s	tar_get_ro_le	breakp_o_int_ge_nomic	break_p_o_int_exo_n	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_do_mains	lost_dom_ains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 5552-T01-I M6-51	P-0035 552-T0 1-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thr_ee _pr im e	886512 51	12	466	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12020963::chr15:g.88651251) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-003 5767-T01-I M6-52	P-0035 767-T0 1-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thr_ee _pr im e	885057 87	14	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143::chr15:g.88505787) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-000 7772-T01-I M5-53	P-0007 772-T0 1-IM5		NTR K3-E TV6	ETV 6	in-fr ame	thr_ee _pr im e	886008 44	13	466	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-000 6480-T01-I M5-54	P-0006 480-T0 1-IM5		NTR K3-E TV6	ETV 6	in-fr ame	thr_ee _pr im e	884981 45	14	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029548::chr15:g.88498145) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-000 2978-T04-I M5-55	P-0002 978-T0 4-IM5		NTR K3-E TV6	ETV 6	in-fr ame	thr_ee _pr im e	885065 71	14	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-000 2978-T05-I M6-56	P-0002 978-T0 5-IM6		NTR K3-E TV6	ETV 6	in-fr ame	thr_ee _pr im e	885065 71	14	529	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) Protein fusion: in frame (ETV6-NTRK3)	NA

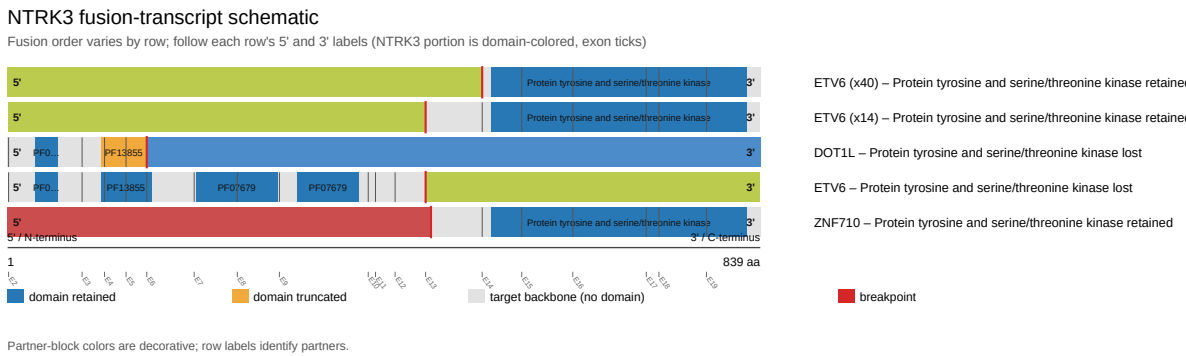
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004737-T01-IM5-57	P-0004737-T01-IM5		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88632786	12	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 13 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12006809::chr15:g.88632786) Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0023257-T01-IM6-58	P-0023257-T01-IM6		NTRK3-ETV6	ETV6	in-frame	threonine_prime	88662190	12	466	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 14 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12014998::chr15:g.88662190) Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0010556-T01-IM5-70	P-0010556-T01-IM5		NTRK3-ZNF710	ZNF710	unknown	threonine_prime	88576258	13	472	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NTRK3 (NM_001012338) - ZNF710 (NM_198526) rearrangement: c.-28-10227:ZNF710_c.1415:NTRK3inv Protein fusion: mid-exon (ZNF710-NTRK3)	NA

Figures

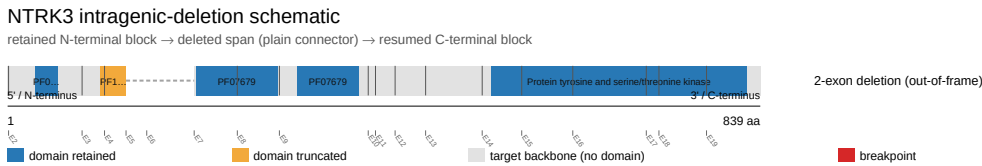
domain retention outliers



fusion schematic



intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

